BOWEN WEI | CV

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Research Interest

My research sits at the intersection of **trustworthy AI** and **LLM reasoning**, with a growing focus on **agentic AI safety** and **reinforcement learning** for building reliable, autonomous systems. I develop **interpretable learning methods**—including **prototype-based** and **neuro-symbolic** approaches—that make model decisions transparent, auditable, and robust for high-stakes applications, revealing what evidence or rules drive a prediction and when the model may fail.

In parallel, I investigate **agent safety evaluation** and **reinforcement learning techniques** for agentic reasoning, including **safety benchmarking under adversarial prompt injection** and **self-play frameworks**. This line of work aims to equip LLM-based agents with the capacity for **verifiable, multi-step reasoning** and principled exploration under uncertainty. Ultimately, I aim to build AI systems that are both **interpretable** in their inner logic and **safe and reliable** in autonomous, evidence-grounded decision-making.

Education

- Ph.D. in Computer Science, George Mason University, Fairfax, VA (Expected graduation: as early as Sep 2027)
- M.S. in Computer Science, University of Virginia (2021–2023)
- B.S. in Computer Science, Xidian University (2016–2021)

Publications

- **UNDER REVIEW AT ICLR 2027** **Bowen Wei**, Han Xie, Jianfeng He, Soumya Sanyal, Eti Rastogi, Yunyi Zhang, Zheng Qi, Jonathan Wu.
HILT: Hazard-Informed Localized Teaching for Safer Language Models.

- **UNDER SUBMISSION** **Bowen Wei**, Yunbei Zhang, Jinhao Pan, Kai Mei, Xiao Wang, Jihun Hamm, Ziwei Zhu, Yingqiang Ge.
ClawSafety: "Safe" LLMs, Unsafe Agents.

- **UNDER SUBMISSION** **Bowen Wei**, Ziwei Zhu.
Neural Symbolic Logical Rule Learner for Interpretable Learning.

- **EMNLP 2026** Main **Bowen Wei**, Nan Wang, Yuqing Zhou, Jinhao Pan, Ziwei Zhu.
COSE: Confidence-Orchestrated Self-Evolution for Effective LLM Reasoning.

- **IEEE BIGDATA 2026** **Bowen Wei***, Jinhao Pan*, Yuqing Zhou, Ziwei Zhu.
A Logical-Rule Autoencoder for Interpretable Recommendations. * Equal contribution. Acceptance: 201 / 1,046 ≈ 19.2%.

- **AAAI 2026** Oral **Bowen Wei**, Mehrdad Fazli, Ziwei Zhu.
Making Sense of LLM Decisions: A Prototype-based Framework for Explainable Classification. Acceptance: 4,176 / 23,680 ≈ 17.6%.

- **ACL 2025** Main **Bowen Wei**, Ziwei Zhu.
ProtoLens: Advancing Prototype Learning for Fine-Grained Interpretability in Text Classification. Acceptance: 1,699 / 8,360 ≈ 20.3%.

- **NEURIPS LAW 2025** **Bowen Wei**, Yuan Shen Tay, Howard Liu, Jinhao Pan, Kun Luo, Ziwei Zhu, Chris Jordan.
CORTEX: Collaborative LLM Agents for High-Stakes Alert Triage.

- **M.SC. THESIS** **Bowen Wei**, Yiling Jia, Hongning Wang.
An Empirical Study of Neural Contextual Bandit Algorithms.

- **ACL 2026** Oral Chahat Raj, **Bowen Wei**, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu.
VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models.

- **ACL 2026** Findings Mehrdad Fazli, **Bowen Wei**, Ziwei Zhu.
Context-Aware Decoding for Faithful Vision-Language Generation.

- **ICML 2026** Jinhao Pan, Chahat Raj, Anjishnu Mukherjee, Sina Mansouri, **Bowen Wei**, Shloka Yada, Ziwei Zhu.
Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement.

- **WACV 2026** Mehrdad Fazli, **Bowen Wei**, Ahmet Sari, Ziwei Zhu.
CAAC: Confidence-Aware Attention Calibration to Reduce Hallucinations in Large Vision-Language Models.

Research Projects

Agentic AI Safety

- **HILT: Hazard-Informed Localized Teaching for Safer Language Models**

  Language-model agents can complete a task while violating its safety constraints through a sequence of individually plausible actions. Developed during my Summer 2026 internship with Amazon Bedrock Guardrails, this project introduces **HILT**, a training framework that uses safety verifiers to locate the *recoverable window* between hazard onset and irreversible commitment. Rather than relying on poorly calibrated absolute verifier scores, HILT measures *Hazard Information* as the change in each verifier's assessment between consecutive trajectory prefixes. Prospective-risk signals estimate when the window opens, while realized-effect signals provide evidence for when recovery may no longer be possible.

  A privileged teacher repairs the trajectory at the estimated opening, and an unprivileged student distills the repair on policy; repairs that occur after the estimated closing boundary are rejected. The resulting policy requires no safety verifier at inference. Across 1,054 harmful agent trajectories, panel-based Hazard Information localizes the trigger with 0.717 exact and 0.786 within-one-step accuracy, while the estimated opening precedes true irreversible commitment in 93.1% of trajectories. Under review at ICLR 2027.

- **ClawSafety: Benchmarking Personal AI Agent Safety Under Prompt Injection**

  Personal AI agents like OpenClaw operate with elevated privileges—accessing local files, email, cryptocurrency wallets, and code execution tools—where a single successful prompt injection can cascade into real-world harm far beyond what chat-level jailbreaks cause. We introduce ClawSafety, a benchmark of 120 adversarial test cases organized along three dimensions: *harm domain* (software engineering, finance, healthcare, law, DevOps), *attack vector* (skill injection, email injection, web injection), and *harmful action type* (data exfiltration, config modification, destination substitution, credential forwarding, destructive action). Each test case is grounded in a realistic, high-privilege professional workspace.

  We evaluate five frontier LLMs across 2,520 sandboxed trials. Attack-success rates range from 40% to 75%, with skill injection consistently presenting the greatest risk. Cross-scaffold experiments further show that agent architecture materially affects safety outcomes. Under submission.

- **CORTEX: Collaborative Multi-Agent LLM Architecture for SOC Alert Triage**

  Security Operations Centers face severe alert fatigue, with false-positive rates approaching 99% and tens of thousands of daily alerts. We propose CORTEX, a divide-and-conquer multi-agent architecture in which specialized agents assume distinct roles: a Behavior Analysis Agent routes alerts to relevant investigative workflows, Evidence Acquisition Agents query enterprise systems, and a Reasoning & Coordination Agent synthesizes the findings into auditable, schema-valid triage reports.

  CORTEX improves actionable F1 from 0.66 to 0.78 and reduces false positives from 24.9% to 14.2% over the strongest evaluated single-agent baseline. Published at the NeurIPS 2025 LAW Workshop.

LLM Reasoning & Self-Evolution

- **COSE: Confidence-Orchestrated Self-Evolution for Effective LLM Reasoning**

  Self-evolving language models reduce reliance on human-curated supervision by generating their own training tasks, solutions, and feedback. However, the same model must often validate generated tasks and judge proposed answers, allowing incorrect self-judgments to become incorrect gradient updates. We introduce **COSE**, a self-evolution framework that treats the model's intrinsic confidence as a lightweight uncertainty signal for controlling how strongly self-generated feedback influences learning.

  COSE combines *confidence-weighted PPO*, which scales policy updates according to feedback reliability, with *confidence-prioritized replay*, which focuses training on informative and trustworthy self-generated experiences. Across 19 held-out benchmarks and four Qwen/Llama backbones ranging from 0.6B to 4B parameters, COSE consistently improves over the corresponding base models, achieves the strongest average performance among evaluated methods on general reasoning and mathematics, and remains competitive on code. Accepted at EMNLP 2026 Main Conference.

Prototype-Based Interpretability

- **ProtoSurE: Prototype-Based Surrogate Explanations for LLM Decisions**

  Existing explanation methods for LLMs either lack faithfulness to the model's actual reasoning or produce token-level attributions that humans find difficult to interpret. We propose ProtoSurE, a framework that trains an interpretable-by-design surrogate model to approximate a target black-box LLM using sentence-level prototypes as human-understandable concepts. The surrogate uses knowledge distillation, attribution-aware sentence embeddings, prototype matching, and diversity regularization.

  ProtoSurE achieves the strongest average faithfulness ranking across four target LLMs and four datasets, outperforming SHAP, Integrated Gradients, Occlusion, and DeepLIFT on six faithfulness metrics. Published at AAAI 2026 as an oral presentation.

- **ProtoLens: Fine-Grained Prototype Learning for Text Classification**

  ProtoLens introduces two core innovations: a *Prototype-aware Span Extraction* module that identifies flexible-length text spans relevant to each prototype and a *Prototype Alignment* mechanism that adaptively maps learned prototype embeddings to representative training examples.

  ProtoLens provides two levels of interpretability: prototypes aligned with human-understandable concepts reveal *what* the model detects, while extracted input spans show *where* each concept appears in the text. Published at ACL 2025 Main Conference.

Neuro-Symbolic & Rule-Based Learning

- **LIA: Logical-Rule Interpretable Autoencoder for Recommendations**

  LIA is a neuro-symbolic autoencoder that produces human-readable logical rules from user behavior. The architecture introduces learnable

AND/OR gates, signed-weight negation, and a disjunctive aggregation layer that combines rule activations into final recommendation scores. The resulting rules are fully inspectable, allowing practitioners to read and audit the reasoning behind recommendations. Accepted at IEEE BigData 2026.

- **NFRL: Neural Symbolic Logical Rule Learner**
  NFRL introduces a selective discrete neural network for learning interpretable classification rules in both conjunctive and disjunctive normal forms. Adaptable AND/OR neurons, explicit negation, and constrained connectivity produce compact, human-readable rules while preserving competitive predictive performance. In progress.

Industry Experience

Applied Scientist Intern - Amazon Web Services (AWS), Seattle, WA — May 2026 – Aug 2026

• Developed **HILT** with the **Amazon Bedrock Guardrails** team, converting temporal changes in safety-verifier signals into localized supervision for safer foundation-model agents

• Designed recoverable-window localization, privileged trajectory repair, and on-policy distillation to produce a student policy that requires no safety verifier at inference

• Evaluated HILT on 1,054 harmful agent trajectories, achieving 0.717 exact and 0.786 within-one-step trigger-localization accuracy, with the estimated intervention point preceding irreversible commitment in 93.1% of trajectories

• Submitted ***HILT: Hazard-Informed Localized Teaching for Safer Language Models*** to ICLR 2027

AI Agents Developer - Fluency Security — Jun 2025 – Aug 2025

• Proposed and built **CORTEX**, a role-specialized multi-agent LLM architecture for SOC alert triage

• Work accepted to the NeurIPS 2025 LAW Workshop as ***CORTEX: Collaborative LLM Agents for High-Stakes Alert Triage***

GenAI Engineer - GoEngage — Jun 2025 – Aug 2025

• Implemented a semantic search engine that improved retrieval accuracy over keyword matching

Professional Service

• **Reviewer** / **Sub-reviewer:** COLM 2026, ICML 2026, NeurIPS 2026, ARR (Dec 2024; Feb 2025—ACL; May 2025—EMNLP), KDD 2024, ACML 2024–2025, SSCI 2025, CAIS.